SUBMISSION COVER LETTER

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Manuscript Title: AETHER-64: High-Throughput Binary Devirtualization via Bit-Parallel Algebraic Pre-Reduction and SSA Bytecode Promotion

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Dear Editor-in-Chief and Editorial Board,

I am pleased to submit our original research manuscript titled **"AETHER-64: High-Throughput Binary Devirtualization via Bit-Parallel Algebraic Pre-Reduction and SSA Bytecode Promotion"** for publication consideration in the **Journal of Computer Virology and Hacking Techniques**.

Alignment with Journal Scope and Context: Commercial virtualization packers (e.g., VMProtect 3.4/3.5, Themida 3.1) and complex compiler obfuscators (Tigress) represent the primary defensive evasion technologies deployed by advanced malware authors, evasive ransomware strains, and state-sponsored APT campaigns to blind endpoint security analyzers and automated sandbox environments. By dismantling native machine code into randomized bytecodes executed by an embedded interpreter, these tools conceal malicious payloads.

Existing reverse engineering frameworks (e.g., angr, Triton) suffer from severe path explosion and SMT solver bottlenecks when analyzing virtualized binaries. In harmony with the **Journal of Computer Virology and Hacking Techniques**' mission to foster rigorous scientific methods addressing real-world hacking techniques, AETHER-64 introduces:

1. **Local-Context Handler Symbolic Slicing:** Traverses complete virtual bytecode control-flow graphs without accumulating path constraints, eliminating solver-induced path explosion.
2. **Bit-Parallel Algebraic Pre-Reduction (APR) Filter:** Employs Fast Moebius Inversion over $\frac{\mathbb{Z}}{2^{64}}\mathbb{Z}$ to resolve 93.91% of opaque branch predicates in $O(1)$ time, yielding an empirical **2,072.8x speedup** over standard SMT solving.
3. **Compiler-Grade SSA Virtual Stack Elimination:** Promotes interpreter memory accesses into clean LLVM 18 SSA registers (`-mem2reg`), stripping 91.8% of instruction bloat.
4. **Practical Validation on Evasive Commercial Protections:** Successfully unpacks and devirtualizes binaries hardened with VMProtect 3.5.1 (defeating inline direct syscalls and FLS heap evasion) and Themida 3.1.2.

Publication Declarations: This manuscript represents original work that has not been published previously and is not under consideration elsewhere. The author declares no conflicts of interest. All evaluation datasets and verification pipelines are openly accessible in an anonymized repository for double-blind peer review.

Thank you very much for your time and editorial consideration.

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